

Singularities — Index & Poincaré-Hopf

$$\text{index}(p) = \frac{1}{2\pi} \oint_{\partial B(p)} d\theta$$

- **4 tip singularities**, each with index $-\frac{1}{4}$
- **Poincaré-Hopf**: sum of indices equals Euler characteristic χ
- **Periodic cylinder**: $\chi = 0$, so indices must balance

$$\sum_i \text{index}(p_i) = \chi(\Omega)$$

Detection: `tools/singularity_detector.py` *Lit*: Kowalski 2015, “Blocking piecewise-linear domains”